MAAI 902 · NLP SOLUTION DESIGN · ASSESSMENT 2 · SUPPLEMENTARY SHEET FOR THE MARKER

Negative-Space Reader
detecting what a document conspicuously omits by modelling what its peers assert

Parsa Hosseini · University of Wollongong in Dubai · October 2026 · Report, repository and live demo submitted together

REPOSITORY · TAG V0.1-CLASS

Code, README, plan, changelog

`github.com/parsa-hke/negative-space-reader/tree/v0.1-class`

Public. Apache 2.0. The tag is the submitted state; later commits on `main` are post-submission.

LIVE DEMO · HUGGING FACE SPACE

Working prototype, no install

`huggingface.co/spaces/parsa-hke/negative-space-reader`

Free CPU hardware. If it is asleep, the first visit wakes it in about a minute.

TECHNICAL REPORT

Submitted on the portal

`…/blob/v0.1-class/reports/report.md`

Same text as the submitted file, with the source and every table it is built from alongside.

The problem. Given a document and a corpus of its genre, rank the kinds of statement the document *lacks* by how many of its 200 closest peers make them, and show the peers' own sentences as evidence. One engine, no genre-specific code: a genre is a **pack** of configuration and data. Two packs ship, privacy policies and cs.CL/cs.LG papers. Nothing is a verdict: every number is a count of peers, every quote is theirs.

2 packs
58 + 47 STATEMENT TYPES

46k documents
28,440 POLICIES · 17,644 PAPERS

2 benchmarks
CONTROLLED DELETION · AUTHORS' OWN REVISIONS

491 tests
PASS IN ≈ 25 S ON A CLEAN CLONE

WHERE TO LOOK IN THE REPOSITORY

`README.md`	Setup, what a fresh clone can run, each pipeline step as a recorded run with duration and hardware.
`NEGATIVE_SPACE_PLAN.md`	The phased plan, each gate ticked and dated; frozen decisions in §0.2.
`CHANGELOG.md`	Dated record of what was done and found, negative findings included.
`engine/` · `packs/`	The genre-agnostic engine; the pack builder and both packs.
`experiments/` · `configs/`	One entry point for every run, one YAML per experiment.
`prompts/`	Every teacher and baseline prompt, versioned and hashed.
`analysis/` · `reports/`	Scripts that regenerate every table and figure; the report is built from them. Decision records in reports/decisions/.
`app/` · `tests/`	The Gradio demo; the test suite.

RUN IT YOURSELF · THREE COMMANDS

1. Install into a virtual environment (Python 3.11).

```
python3.11 -m venv .venv
.venv/bin/pip install -e .
```

2. Run the tests. They need no data and no large model weights.

```
.venv/bin/python -m pytest -q   # 491 passed
```

3. Build a pack in smoke mode on the 200-document corpus (README explains the data download).

```
python -m experiments.run \
    --config configs/pack_privacy.yaml \
    --overlay local --overlay smoke
```

HEADLINE RESULT · OMITBENCH-SYNTHETIC, RECALL@5

PAPERS · 500 INSTANCES



PRIVACY POLICIES · 500 INSTANCES



Single-type deletion, one hidden statement per document. Peers beat genre-wide frequency for papers and not for policies; the report's Section 6 explains why. The chat-model rows are the 100-instance frontier comparison (Table 8), reported with confidence intervals there.

WHAT KEEPS THE NUMBERS HONEST

- Every experiment runs through `experiments/run.py` with a config; each run records its config hash, git commit, device and the version and hash of every prompt it rendered.
- The report's source carries no typed number: every value is filled by key from `reports/tables/report_fragments/values.json`, written by `analysis/tables.py` from the run outputs.
- Tests assert that the committed report, tables and figures equal a fresh build, so the submitted text cannot drift from the results.
- Smoke mode (200 documents per genre, small models) must pass before any full run; the fresh-clone test of 2026-10-25 is recorded in the changelog.
- Negative findings are kept: the plan and changelog record what did not work and why, including the privacy-policy result above.

USING THE DEMO

- Start with a **pinned example**: the guided walkthrough narrates each step and ends with a challenge-and-reveal on a benchmark instance.
- Paste or upload your own policy or paper; auto-detect picks the pack. On the free CPU a policy takes about 10 s and a paper about 36 s.
- Click a **ghost paragraph** to open its evidence: peer count, five peer sentences with document ids, suggested wording.
- The cohort panel re-ranks without re-embedding: change k, the category filter or the date window and watch the list move.
- The **About** tab carries the same numbers as the report tables.

MODELS AND DATA, FOR THE RECORD

`Schema induction`	Qwen2.5-32B (lab), confirmed on Qwen2.5-72B
`Teacher labels`	Qwen2.5-72B-Instruct; silver reference by Claude Haiku 5.5
`Embeddings`	Qwen3-Embedding-0.6B (packs); bge-small in smoke
`Frontier comparison`	Claude Opus 5.5 and Qwen2.5-72B, candidate-list setting
`Corpora`	Privacy policies 1998–2019; permissively licensed arXiv cs.CL/cs.LG 2018–2022

Parsa Hosseini · GitHub `parsa-hke` · Hugging Face `parsa-hke`

Repository at `v0.1-class` · Space `parsa-hke/negative-space-reader`